

Nuclear prep and FACS for 10X Genomics Chromium single-nucleus sequencing, V1.3

Partially from Habib et al., *Massively parallel single-nucleus rna-seq with dronc-seq*, Nature Methods 2017. (With additional commentary by Eugene Drokhlyansky and Ehsan Habibi, Regev lab. Per Gene, this method for nuclear extraction works best. Per Ehsan, this method works very well for brain)

Reagents:

EZ Lysis Buffer from Nuclei EZ Prep kit (Sigma, Cat # NUC-101)

* ensure that the lysis buffer is the version designed for cell culture, and not the version designed for tissue

2ml glass dounce tissue grinder (e.g., Sigma, Cat # D8938)

* if using a different product, ensure pestles have ball end and not U-end, or they will damage the nuclei

35- μ m cell strainer (e.g., Corning, Cat # 352235)

1ml Pipettor and pipette tips (note: 2ml serological pipettes do not fit in the dounce)

Nuclei Suspension Buffer (NSB):

1 $\times$ RNase-free molecular-biology grade PBS

0.1% molecular-biology grade BSA (100 μ g/ml) (e.g., NEB B9000S)

0.2U/ μ l RNase inhibitor (e.g., Clontech, Cat # 2313A)

For 5 ml:

5 ml PBS

25 μ l NEB BSA (@ 20 mg/ml)

25 μ l RNase inhibitor (@ 40U/ μ l)

For 2 samples $\rightarrow$ 12 ml

PBS 12 ml

BSA 60 μ l

RRi 60 μ l

mix well by inverting and take 3 ml for FACS

3 ml for FACS

Add 6 μ l Ruby (see below) 15 μ l RRi

3 ml PBS
15 μ l BSA
30 μ l RRi
6 μ l Ruby

9 ml PBS
45 μ l BSA
45 μ l RRi

Tissue harvesting and storage:

Dissect tissue and place in 1.5 ml microcentrifuge tube, transferring as little media as possible. If necessary, cut tissue samples into pieces <0.5 cm first. Flash-freeze on dry ice and store at -80°C.

Gene says: Dissect the region of interest, place on the inside wall of a 1.5 ml tube, make sure it doesn't slide down, and freeze tube in dry ice. The sliding part is to make sure there is no excess liquid on the tissue. You don't want to pat dry, but you don't want tissue to freeze in any liquid.

Nuclear prep:

* Take a drop of the 3 ml NSB in an eppendorf at RT and add the 6 μ l Ruby. Mix well and transfer all to the Falcon tube

Note: Maintain RNase free conditions throughout. Keep nuclei ice-cold at all times.

1. Preparation:

- Pre-chill centrifuge.
- Place a 15-ml conical tube on ice for each sample.
- Wash dounce and pestles with 100% EtOH, followed by RNase Zap. Wash in 2-3 rounds of RNase-free water (in 50 ml conical), and rinse with EZ Lysis Buffer.
- Chill dounce and pestles on ice (pestles in a 50 ml conical tube).
- Prepare ~5 ml Nuclei Suspension Buffer per sample (plus ~1-2 ml extra if doing FACS) and place on ice.

- f. If you are using the Vybrant DyeCycle Ruby stain, make ~1 ml NSB & Ruby (1:500) per sample (or a bit extra if you think you might need to dilute the sample). Add an additional 5 μ l / ml RNase inhibitor to this solution (final 0.4U/ μ l), because the nuclei will be sitting in it for a while.
 - The Ruby is provided in a DMSO solution and is not very soluble in cold media; if necessary, dilute it 1/10 in a small amount of room-temperature NSB, then 1/50 into the cold NSB (1:500 final).
- g. Prepare ~100 μ l of rich-NSB to sort the cells into: 100 μ l NSB (without Ruby) with 5 μ l extra RNase inhibitor (final 2U/ μ l) and 5 μ l extra BSA (final 1%).
- h. Plan to sort into a 96-well plate; pick one well per sample and coat the entire inside with ~20 μ l of the NEB BSA (undiluted). Place the plate in a clean ziploc bag to transport it to/from the core. (The BSA coating step reduces adherence and improves recovery.)
 - Before sorting, draw off the BSA in the wells and rinse the well 1-2 times with ~100 μ l NSB (the basic one), then fill with ~10-20 μ l of rich-NSB (based on the volume you expect to collect – try to keep the final post-sort volume under 30 μ l). Try to make sure there are no bubbles.

Paul uses wells
D6 & D9

2. Transfer tissue to chilled dounce. Do not allow tissue to thaw. For pieces of mouse cortex under 0.5 cm, it is not necessary to chop the tissue. Immediately add 2 ml ice-cold EZ Lysis Buffer.
 3. Homogenize with 20 strokes of pestle A and 20 strokes with pestle B. Raise and lower slowly, avoiding bubbles. Keep nuclei on ice at all times.
 4. Transfer suspension to chilled conical (from step 2). Rinse dounce with 2 ml ice-cold EZ Lysis Buffer; transfer rinsate to conical and mix gently. Incubate on ice 5 min.
 5. Centrifuge at 500 $\times$ g for 5 min at 4°C.
 6. Resuspend in 4 ml ice-cold EZ Lysis Buffer by gentle pipetting and incubate on ice for 5 min.
 7. Centrifuge at 500 $\times$ g for 5 min at 4°C.
 8. Resuspend in 4 ml ice-cold Nuclei Suspension Buffer.
 9. Centrifuge at 500 $\times$ g for 5 min at 4°C.
 10. Resuspend in ~1 ml ice-cold Nuclei Suspension Buffer with Ruby (depending on expected yield). Filter through a 35- μ m cell strainer.
 - To count or observe the nuclei, they may be stained with DAPI or Trypan Blue (Ruby is not in the visible spectrum). Nuclear morphology should be relatively rounded; jagged or excessively elongated shapes indicate damage.
- The number of strokes in step 4 and the incubation times in steps 5 and 7 may need optimization based on tissue. If too many nuclei have adherent cell material, try increasing. If there is excessive damage, try decreasing. (See EZ Prep kit documentation for additional troubleshooting.) Clumping may also indicate the need for a higher concentration of BSA in the final resuspension buffer (per 10X Genomics single-nucleus protocol, BSA can go up to ~1%).

FACS:

Be extra thorough with all pre-sort cleaning steps, but usually no other special cleaning procedures are needed (bleach is an excellent anti-RNase treatment). Use only RNase-free water and PBS for all washing/priming steps after the bleach treatment. Try to avoid following after people using serum-containing medium; if you must, clean even more.

Use chilled settings for sample and collection tubes/plates. Pressure should be kept as low as feasible.

For the 10X Genomics Chromium Single Cell System, sort into the 96-well plate prepared above, spin down to reduce volume if necessary, and immediately proceed with the standard 10X Genomics protocol. Note that after breaking the GEMs the aqueous phase may be slightly milky due to the BSA and RNase inhibitor.

- For the cDNA amplification step, calculate based on ~50-60% recovery and then add 1-2 cycles over the recommended number, as nuclei have less RNA than cells.

Insofar as possible, the sorted cells should be collected into a small volume (≤ 30 μ l) to avoid the need to spin down after sorting. We have found that a 100 μ M nozzle leads to a larger collection volume than is desirable (>30 μ l for ~7000 events); if a smaller nozzle (e.g., 85 μ M) gives adequate nuclear quality, it may be preferable.

Further notes from advisors:

Gene recommends using a 100 μ M nozzle and Vybrant DyeCycle Ruby stain (Invitrogen, Cat # V10309), which is a far-red dye chosen for contrast with GFP but is also compatible with RFP/Tomato (conventional wisdom in Regev lab: DAPI inhibits reverse transcriptase and leads to lower quality sequence). Pressure should be about 22-24 (for whatever machine Gene is using)

For in-plate protocols, ensure droplets hit the lysis buffer directly and not the side of the well.

Per Danielle Dione, Regev lab:

For most sample types that we are running nuclei on we have run the exact same protocol as for cells for all of the 10X loading and downstream processing. We generally load 7000 cells/nuclei per channel. There are a few sample types that we have found require a couple extra cycles of PCR during the cDNA amplification but it has not been all samples like we expected. We usually start with 11 cycles of PCR and if we see low quants (<1 ng/ μ l) we increase for future samples.